Judicial Digital Support Platform

Software Requirements Specification (SRS):

1.Introduction:

- **1.1 Purpose:**
  - This document defines the software requirements for the Judicial Digital Support Platform (JDSP).
  - The system aims to digitally support judicial processes by improving efficiency, transparency, security, and accessibility while preserving judicial authority and legal compliance.
- **1.2 Scope:**
  - The JDSP is a web-based platform designed to assist courts in **managing case workflows**, **digital documentation**, **payment processing**, **notifications**, **controlled online hearings, Lawyer Roll Tracking, Lawyer Roll Tracking.**
- **1.3 Definitions:**
  - JDSP – Judicial Digital Support Platform.
  - RBAC – Role-Based Access Control.
  - Audit Log – Immutable activity tracking record.

2.Problem Statement:

- The current judicial workflow relies heavily on manual procedures, physical attendance, and paper-based documentation.
- This results in:
  - Delays in case processing
  - Administrative inefficiencies
  - Limited transparency for litigants
  - Increased operational costs
  - Difficulty tracking case status
- There is a need for a secure digital support platform that enhances judicial processes without replacing the legal authority of courts.

3.Target Users:

- Magistrate
- Lawyer
- Client
- court employee
- Senior management

4.System Scope:

- **4.1 In Scope:**
  - ❖ Administrative analytics dashboard (Case Tracking Dashboard)
  - ❖ Role-Based Access control
  - ❖ Case registration and tracking
  - ❖ Realtime messaging
  - ❖ Electronic Payments (local payment (ex. Vodafone Cash))
  - ❖ Notifications (SMS + Email + application)
  - ❖ Document Upload & Storage
  - ❖ Audit Loging system
    (It is an immutable log that records every operation that occurs in the system:
    - Who did what?
    - When?
    - And from where (IP/Device)?
  - ❖ User authentication (2FA Authentication)
  - ❖ Online Hearings (for selected case types only)
  - ❖ Hearing scheduling
  - ❖ Lawyer Roll Tracking
  - ❖ Temporary Location Permission
- **4.2 Out of Scope:**
  - ❖ Issue automated judicial rulings.
  - ❖ Replace physical court proceedings entirely.
  - ❖ Handle high-risk criminal case trials.
  - ❖ Make AI-based legal decisions.

5. Functional Requirements:

- **5.1 Authentication & Authorization:**
  - ❖ The system shall implement 2FA for sensitive roles.
  - ❖ The system shall enforce RBAC.
  - ❖ The system shall log all login attempts.
- **5.2 Case Management:**
  - ❖ The system shall allow lawyers to submit new cases.
  - ❖ The system shall allow court employees to review submissions
  - ❖ The system shall generate a unique case ID.
  - ❖ The system shall track case status.

- **5.3 Hearing Management:**
  - ❖ The system shall allow scheduling of hearings.
  - ❖ The system shall support online hearings for approved case types.
  - ❖ The system shall record hearing sessions.
- **5.4 Payment System:**
  - ❖ The system shall integrate with payment gateways.
    (Digital wallet (ex. Vodafone cash, bank transfers, payment gateways (ex. fawry))
  - ❖ Each payment shall be linked to a case ID.
  - ❖ The system shall generate digital receipts.
- **5.5 Notification System:**
  - ❖ The system shall send SMS notifications.
  - ❖ The system shall send email notifications.
  - ❖ The system shall notify users of status changes.
- **5.6 Document Management:**
  - ❖ The system shall allow uploading legal documents.
  - ❖ Documents shall be stored securely.
  - ❖ The system shall record document upload history.
- **5.7 Audit Logging:**
  - ❖ The system shall log all create, update, delete, and view actions.
  - ❖ Audit logs shall be immutable.
  - ❖ Audit logs shall record user ID, timestamp, action type, and entity affected.
- **5.8 Real-Time Communication:**
  - ❖ The system shall provide chat between lawyer and client.

6.Non-Fuctional Requirements:

- **6.1 Security:**
  - ❖ All communications shall use HTTPS.
  - ❖ Sensitive data shall be encrypted at rest.
  - ❖ The system shall enforce 2FA for judges and court employees.
  - ❖ The system shall implement audit logging.
- **6.2 Performance:**
  - ❖ The system shall support concurrent users (target defined in deployment).
- **6.3 Availability:**
  - ❖ The system shall maintain 99% uptime.

- ❖ The system shall perform automatic daily backups.

- **6.4 Scalability:**
  - ❖ **The system shall support expansion to multiple courts.**
  - ❖ **The architecture shall allow modular scaling.**
- **6.5 Usability:**
  - ❖ The UI shall be simple and accessible.
  - ❖ The system shall support Arabic (primary) and English (secondary).
  - ❖ The platform shall be accessible to non-technical users.

7.Constraints:

- Legal compliance with national judicial regulations.
- Data privacy laws.
- Government hosting requirements.
- Internet connectivity limitations.
- Budget constraints.
- Resistance to institutional change.

8.Use Cases (High-Level):

- **UC-01: User Login**

  Any User:

  1. User enters credentials.
  2. System validates credentials.
  3. System requests 2FA.
  4. User gains access based on role.
- **UC-02: Submit New Case**

  Lawyer:

  1. Lawyer uploads documents.
  2. Lawyer fills case form.
  3. System validates input.
  4. Case ID generated.
  5. Court employee notified.
- **UC-03: Schedule Hearing**

  Court Employee:

  1. Select case.

2. Choose date
3. Define hearing type (online/Physical)
4. Notify involved parties.

- **UC-04: Make Payment**

  Lawyer / Client:

  1. Select case.
  2. Choose payment method.
  3. Complete transaction.
  4. Receipt generated.